Data Exploration

Project Objective:

This project is part of the Data Analytics (DA) Module – Excel, focusing on foundational data exploration techniques using Microsoft Excel. The goal of this project is to apply Excel formulas to summarize data, categorize values, extract text, and perform conditional calculations.

Dataset Description:

Focuses on Excel-based data exploration and analysis using a structured product dataset. The goal is to build foundational skills in data summarization, logical functions, conditional aggregation, and text manipulation—all essential techniques for preparing data for deeper analytics.

This dataset containing product details such as Product ID, Product Name, Brand, Category, Quantity, and Price. Using Excel formulas and perform a series of analytical tasks to extract insights, categorize data, and transform text fields.

Column Description:

Column Name	**Description**
Product ID	ID Numbers
Product Name	Name of the Product
Brand Name	Brand name of Product
Price	Price of the Product
Quantity	No of Quantity by Product
Category	Category based on the Product

Data cleaning & transformation:

1. Basic Data Exploration

- ✓ Total price of all products
- ✓ Total number of products
- ✓ Average price of products

Formulas Used:

=SUM(D2:D35)
=COUNT(B2:B35)
=AVERAGE(D2:D35)

2. Minimum & Maximum Price

- ✓ Identified lowest product price
- ✓ Identified highest product price

Formulas Used:

=MIN(D2:D35)
=MAX(D2:D35)

3. Logical Function – IF

- ✓ Created a new column Price Range:
    - ♦ Price ≥ 500 → High Price
    - ♦ Price < 500 → Standard Price

Formulas Used:

=IF(D2>=500, "High Price", "Standard Price")
=IFS(D2>=500, "High Price", D2<500, "Standard Price")

4. Conditional Functions – SUMIF & COUNTIF

- ✓ Total price of Electronics category
- ✓ Count of products priced below $100

Formulas Used:

=SUMIF(F2:F35, "Electronics", D2:D35)
=COUNTIF(D2:D35, "<100")

5. Text Extraction – LEFT, RIGHT, MID

- ✓ Created new columns from Product ID

Formulas Used:

Day =LEFT(A2,2)
Country Code =RIGHT(A2,2)
Month =MID(A2,4,3)

Conclusion:

This project demonstrates essential Excel skills required for data analysis. The completed workbook and documentation show correct application of formulas, structured data processing, and clear analytical thinking.